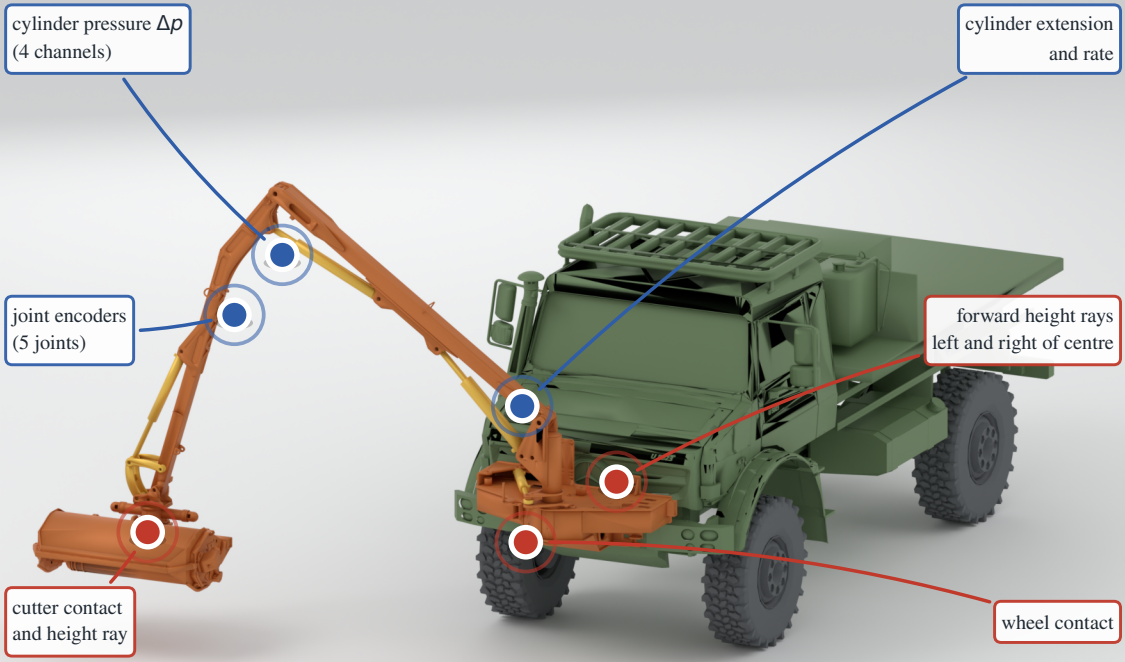

cylinder pressure Δp
(4 channels)
cylinder extension
and rate
joint encoders
(5 joints)
forward height rays
left and right of centre
cutter contact
and height ray
wheel contact
exteroceptive — world contact
proprioceptive — machine state